

CURRICULUM VITAE

Varga Kalantarov, DSc, PhD

Professor, Department of Mathematics

Koç University, Istanbul, Turkey

Contact and identifiers

Full name (as in publications)	Varga K. Kalantarov
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Scientific field	Partial differential equations

Profile summary

Varga Kalantarov is professor in the Department of Mathematics at Koç University in Istanbul, where he has worked since 2001. He received a Doctor of Sciences in differential equations and mathematical physics from the V. A. Steklov Institute of Mathematics at St. Petersburg in 1988. The dissertation is Global behavior of solutions of nonlinear equations of mathematical physics of classical and non-classical types. He received a PhD in the same field in 1974 from the Institute of Mathematics and Mechanics of the Academy of Sciences of Azerbaijan. The dissertation is On mixed problem for semi-linear parabolic-hyperbolic systems. He studied at Azerbaijan State University in 1966–1971. From 1989 to 1993 he headed the Department of Partial Differential Equations at the Institute of Mathematics and Mechanics in Baku and was deputy director of the institute in 1984–1993.

Education

Years	Degree	Field / thesis title	Institution, city, country
1988	DSc	Differential equations and mathematical physics. Dissertation: Global behavior of solutions of nonlinear equations of mathematical physics of classical and non-classical types	V. A. Steklov Institute of Mathematics, St. Petersburg, Russia
1974	PhD	Differential equations and mathematical physics. Dissertation: On mixed problem for semi-linear parabolic-hyperbolic systems	Institute of Mathematics and Mechanics, Academy of Sciences of Azerbaijan, Baku
1966–1971	University study		Azerbaijan State University, Baku, Azerbaijan

Academic and professional positions

Years	Position	Department / unit	Institution, city, country
2001–present	Professor		Department of Mathematics, Koç University, Istanbul, Turkey
1993–2001	Professor		Department of Mathematics, Hacettepe University, Ankara, Turkey

Years	Position	Department / unit	Institution, city, country
1991–1993	Professor, part-time		Department of Applied Mathematics, Baku State University, Baku, Azerbaijan
1989–1991	Professor, part-time		Department of Mathematics, Azerbaijan Institute of Civil Engineering, Baku, Azerbaijan
1989–1993	Head of the Department of Partial Differential Equations		Institute of Mathematics and Mechanics, Baku, Azerbaijan
1984–1993	Deputy director		Institute of Mathematics and Mechanics, Baku, Azerbaijan
1977–1984	Senior scientific researcher		Institute of Mathematics and Mechanics, Baku, Azerbaijan
1974–1977	Scientific researcher		Institute of Mathematics and Mechanics, Baku, Azerbaijan
September 1975–May 1980	Visiting researcher		V. A. Steklov Institute of Mathematics, St. Petersburg
April 2017	Visiting professor		Silkroad Mathematics Center, Beijing, China
January 2017	Visiting professor		Weizmann Institute, Rehovot, Israel
July–August 2015	Visiting professor		University of Toronto, Canada
December 2013–February 2014	Visiting professor		KAUST
July–August 2008	Visiting professor		University of Tennessee, United States
February–June 2001	Research associate		Feza Gürsey Institute of Basic Sciences
November 1997	Visiting professor		Weierstrass Institute for Applied Analysis and Stochastics, Berlin, Germany
September 1994	Visiting professor		Autonomous University of Barcelona, Spain
September–December 1992	Visiting professor		Gazi University, Ankara, Turkey
September 1991	Visiting professor		Polytechnic University of Catalonia, Barcelona, Spain
December 1983	Visiting professor		Karl Weierstrass Institute for Mathematics

Publications

Book chapters

1. **Kalantarov, V. K.** Global solutions of coupled Kuramoto-Sivashinsky and Ginzburg-Landau equations, *Nonlinear problems in mathematical physics and related topics, II*, pp.213–227, Int. Math. Ser. (N. Y.), 2, Kluwer/Plenum, New York, 2002.
2. **Kalantarov, V. K.** Blow-up theorems for second order nonlinear evolutionary equations, in O.Boratav, A.Eden, A.Erzan (Eds.) "Turbulence modeling and vortex dynamics" Lecture Notes in Physics, Springer Verlag, 1997, pp.169-181

Peer-reviewed journal articles

1. **Kalantarov, V. K.** Determining functionals and finite-dimensional reduction for dissipative PDEs revisited. *Journal of Differential Equations*, 345 (2023), 78-103 (with A. Kostienko and S. Zelik)
2. **Kalantarov, V. K.** On blow up of solutions of coupled systems of parabolic and hyperbolic equations. *Math. Notes*, 112:3 (2022), 406–411 (with J. Kalantarova)
3. **Kalantarov, V. K.** Asymptotic regularity and attractors for slightly compressible Brinkman - Forchheimer equations, *Applied Mathematics and Optimization*, 2021 (with S. Zelik)
4. **Kalantarov, V. K.** Blow-up Solutions of Helmholtz Equation for a Kerr Slab with a Complex Linear and Nonlinear Permittivity, *J. Math. Phys.* 60, 043508 (2019) (with A. Mostafazadeh and N. Ofiaz)
5. **Kalantarov, V. K.** Global Stabilization of the Navier-Stokes-Voigt and the damped nonlinear wave equations by finite number of feedback controllers *Discrete and Continuous Dynamical Systems – B*, 23(3) (2018), 1325–1345. (with E.S. Titi)
6. **Kalantarov, V. K.** Existence of an attractor and determining modes for structurally damped nonlinear wave equations, *Physica D, Nonlinear Phenomena*, 376-377(2018) 15-22. (with B.A. Bilgin)
7. **Kalantarov, V. K.** Non-existence of Global Solutions to Nonlinear Wave Equations with Positive Initial, *Communications in Pure and Applied Analysis*, 17(3)(2018) 987-999. (with B.A. Bilgin)
8. **Kalantarov, V. K.** Determining functionals for damped nonlinear wave equations, *Complex Variables and Elliptic Equations*, 63 (2018) 931-944, (with B.A. Bilgin)
9. **Kalantarov, V. K.** On Decay of Solutions of Generalised Pipe Equation Under Nonhomogeneous Dirichlet Boundary Control, *Appl. Comp. Math.*, (2018),17(1), 48-55. (with M.Meyvaci) 4 ===== PAGE 5/8 =====
10. **Kalantarov, V. K.** Preventing blow up by convective terms in dissipative PDE's. *J. Math. fluid Mech.* 18(2016), no. 3, 463-479.(with B. A. Bilgin and S. Zelik)
11. **Kalantarov, V. K.** Attractors of damped quintic wave equations in bounded domains, *Annales Henri Poincaré, A Journal of Theoretical and Mathematical Physics*, Ann. Henri Poincaré 17 (2016), no. 9, 2555-2584.(with A. Savostianov and S. Zelik)
12. **Kalantarov, V. K.** Finite-Parameters Feedback Control for nonlinear Wave Equations, *Contemporary Mathematics*, AMS 269 (2016)(with E. S. Titi)
13. **Kalantarov, V. K.** Qualitative Properties of solutions for nonlinear Schrödinger equations with nonlinear boundary conditions on the half line, *J. Math. Phys.* 57 021511 (2016) (with T. Özşarı)
14. **Kalantarov, V. K.** A note on strongly damped wave equation with fast growing nonlinearities, (with S. Zelik)*Journal of magthem. Phys*,56 ,011501 (2015)
15. **Kalantarov, V. K.** Blow up of solutions to the initial boundary value problem for quasilinear strongly damped wave equations (with Bilgesu A. Bilgin). *J. Math. Anal. Appl.* 403 (2013), no. 1, 89–94.
16. **Kalantarov, V. K.** Smooth attractors for the Brinkman-Forchheimer equations with fast growing nonlinearities. (with S.V. Zelik) *Commun. Pure Appl. Anal.* 11 (2012), no. 5, 2037–2054.
17. **Kalantarov, V. K.** Uniform decay rates for the energy of weakly damped defocusing semilinear Schrödinger equations with inhomogeneous Dirichlet boundary control (with I.Lasiecka and T. Özşarı), *Journal of Differential Equations*, Vol. 251 (2011) 1841 - 1863
18. **Kalantarov, V. K.** Global attractors and determining modes for the 3D Navier–Stokes–Voigt equation (with E.S. Titi) *Chin. Ann. Math. Ser. B* 30 (2009), no. 6, 697–714.
19. **Kalantarov, V. K.** Finite-dimensional attractors for the quasi-linear strongly-damped wave equation (with S. Zelik) *J. Differential Equations* 247 (2009), no. 4, 1120–1155.

20. **Kalantarov, V. K.** Gevrey regularity for the attractor of the 3D Navier-Stokes-Voight equations (with B.Levant and E.S.Titi) *Journal of Nonlinear Science*, 19 (2009) 133 - 152
21. **Kalantarov, V. K.** 3D Convective Cahn - Hilliard Equation, (with A. Eden) *Comm. Pure Appl. Anal.* Vol.6, (2007) no.4, pp. 1075 -1086
22. **Kalantarov, V. K.** On continuous dependence on coefficients of the Brinkman-Forchheimer equations. (with A.O.Çelebi and D.Uğurlu) *Appl. Math. Lett.*, 19 (2006), no. 8, 801–807.
23. **Kalantarov, V. K.** On global behavior of solutions to an inverse problem for nonlinear parabolic equations, (with A.Eden) *J. Math.Anal.Appl.* 307(2005)120-133
24. **Kalantarov, V. K.** The asymptotic behavior of solutions to an inverse problem for differential operator equations. *Math. Comput. Modelling* 37 (2003), no. 9-10, 907–914
25. **Kalantarov, V. K.** A remark on nonexistence of global solutions to quasi-linear hyperbolic and parabolic equations, (with Erdem D.) *Appl. Math. Lett.* vol. 15, no. 5, 521-653 (2002) 5 ===== PAGE 6/8 =====
26. **Kalantarov, V. K.** Spatial behavior estimates for the wave equation under nonlinear boundary conditions, (with A.O. Çelebi) *Math. Comp. Modeling*, 34,(2001),no.5-6.,527-532
27. **Kalantarov, V. K.** Determining functionals for nonlinear damped wave equations, (with Chueshov I.D.) *Math. Phys. Anal. Geom.* 8(2001), no. 215-227.
28. **Kalantarov, V. K.** Attractors for the Generalized Benjamin- Bona -Mahony Equation, (With O. Çelebi ,M.Polat) *Journal of Differential Equations*, 1999,vol.157, pp. 439 - 451.
29. **Kalantarov, V. K.** A remark on two constructions of exponential attractors for a-contractive (with A.Eden and C.Foias) *Journal of Dynamics and Diff.Eq.*,1998, vol.10, no.1, pp.37-45
30. **Kalantarov, V. K.** Finite dimensional attractors for a class of semi-linear wave equations, (with A.Eden) *Turkish Journal of Mathematics*, 1996, v.20, pp. 425-450.
31. **Kalantarov, V. K.** On minimal global attractor for the phase -field equations, *Zap. Nauchn.Sem.. Leningrad Otd.. Mat. Inst. Steklov*, 1991, v.188, no.22, pp.70-86
32. **Kalantarov, V. K.** Global behavior of solutions to some fourth-order nonlinear equations *Zap. Nauchn. Sem.Leningrad Otd.. Mat. Inst. Steklov* 1987, v.1163, no. 19, pp. 6675.
33. **Kalantarov, V. K.** Global behavior of solutions to the Cauchy problem for second order operator- differential equations, *Izv. AN.Azerb.SSR ser.fiz.tech.mat.nauk*, 1987, v.8, no.6, pp. 28-34.
34. **Kalantarov, V. K.** Attractors for some nonlinear problems of mathematical physics *Zap. Nauchn. Sem. Leningrad Otd. Mat. Inst. Steklov* 1986, v.182, pp.50-54. (English translation in *J. Sov. Math.*, 1988, v.40., pp.619-622)
35. **Kalantarov, V. K.** Determination of solutions of the first boundary value problem for a system of Karman equations having an unbounded energy integral, *Zap.Nauchn. Sem. Leningrad Otd.Mat.Inst.Steklov* 1980, v.96, pp.97-100 (English translation in *J.Sov.Math.*, 1983, v.21., pp.711-714)
36. **Kalantarov, V. K.** On stabilisation of solutions of some quasi-linear parabolic equations, (with O. A. Ladyzhenskaya) *Sibirski Mat.Zhurnall*, 1978, v.19, no.5, pp.1043-1052 (English transl. in *Siberian Math J.*, 1978)
37. **Kalantarov, V. K.** The occurrence of collapse for quasi-linear equations of parabolic and hyperbolic type, (with O.A.Ladyzhenskaya)*Zap.Nauchn. Sem. Leningrad Otd.Mat.Inst.Steklov*1977
38. **Kalantarov, V. K.** Investigation of generalized solutions of mixed problem for the system of two equations of parabolic and hyperbolic type, *Uch.. Zap. Azerb.. Gos.. Univ., ser.fiz.mat. nauk*, 1972, no.4, pp. 50-58

Grants and research projects

Years	Project title	Funding body	Your role
2022–2025	Stabilization of infinite-dimensional dissipative systems by finite-dimensional boundary controllers. With T. Özşarı. Grant 112F084	TÜBİTAK	
2013–2014	Blow-up of solutions to damped nonlinear wave equations and nonlinear parabolic equations. Grant 112T914	TÜBİTAK	

Years	Project title	Funding body	Your role
2006–2009	Partial differential equations in phase transitions. With H. M. Soner. Grant 106T337	TÜBİTAK	
1996–1997	Initial-boundary-value problems for the nonlinear Schrödinger equation	TÜBİTAK	

Supervision

Level	Number	Notes
PhD	16	Named theses from 1985 to 2019 at Koç University, Hacettepe University, Middle East Technical University, Sakarya University, Al-Farabi Kazakh National University, and the Institute of Mathematics and Mechanics in Baku. The titles are not printed

Teaching

Years	Course	Level	Institution
2001–present	Calculus, differential equations, multivariable calculus, linear algebra, applied mathematics, partial differential equations, complex analysis, and applied functional analysis. Course numbers include Math 106, 204, 200, 203, 303, 309, 401, 503, 552, 505, 207, and 557		Koç University, Istanbul
	Calculus, ordinary and partial differential equations, functional analysis, Sobolev spaces, nonlinear partial differential equations, blow-up theorems, and dissipative dynamical systems	Graduate and undergraduate	Baku State University, Azerbaijan University of Architecture and Construction, St. Petersburg State University, Ankara University, and Hacettepe University

Invited talks and conference participation

Year	Event, place	Type	Title of talk
	International Conference "O.A. Ladyzhenskaya centennial conference on PDE's", Euler International Mathematical Institute, St. Petersburg, July, 2022.	Selected presentation	International Conference "O.A. Ladyzhenskaya centennial conference on PDE's", Euler International Mathematical Institute, St. Petersburg, July, 2022.
	International Conference "Differential Equations and Dynamical Systems", Silkroad Mathematics Center, Shanghai, March, 2021. 2 ===== PAGE 3/8 =====	Selected presentation	International Conference "Differential Equations and Dynamical Systems", Silkroad Mathematics Center, Shanghai, March, 2021. 2 ===== PAGE 3/8 =====
	International conference "Infinite-dimensional Dynamical Systems and Attractors" Lanzhou University, China, 1-6 July, 2019.	Selected presentation	International conference "Infinite-dimensional Dynamical Systems and Attractors" Lanzhou University, China, 1-6 July, 2019.

Year	Event, place	Type	Title of talk
	International conference “PDE’s and Mathematical Hydrodynamics” St.Petersburg, 30 July - 3 August 2018	Selected presentation	International conference “PDE’s and Mathematical Hydrodynamics” St.Petersburg, 30 July - 3 August 2018
	International conference “Operators, Functions and Systems of Mathematical Physics”, Baku, 21-24 May 2018.	Selected presentation	International conference “Operators, Functions and Systems of Mathematical Physics”, Baku, 21-24 May 2018.
	International workshop “Applied and numerical analysis, with applications”, Bath, July 2016.	Selected presentation	International workshop “Applied and numerical analysis, with applications”, Bath, July 2016.
	International conference “Progress in Nonlinear Partial Differential Equations”, Lisbon, March, 2014,	Selected presentation	International conference “Progress in Nonlinear Partial Differential Equations”, Lisbon, March, 2014,
	International workshop “Advances in Analysis of PDEs” , Stockholm, July, 2012,	Selected presentation	International workshop “Advances in Analysis of PDEs” , Stockholm, July, 2012,
	Dissipative PDEs in Bounded and Unbounded Domains and Related Attractors, Edinburgh, UK, September, 2010	Selected presentation	Dissipative PDEs in Bounded and Unbounded Domains and Related Attractors, Edinburgh, UK, September, 2010
	Third Conference of the World Mathematical Society of Turcic Countries, Almaty, Kazakhstan, July 2009,	Selected presentation	Third Conference of the World Mathematical Society of Turcic Countries, Almaty, Kazakhstan, July 2009,
	Conference on parabolic and Navier – Stokes equations, The Banach Center, Bedlewo, Poland, September, 2008,	Selected presentation	Conference on parabolic and Navier – Stokes equations, The Banach Center, Bedlewo, Poland, September, 2008,
	Annual conference of Turkish mathematical Society, Istanbul, September, 2008,	Selected presentation	Annual conference of Turkish mathematical Society, Istanbul, September, 2008,
	6th International ISAAC Congress, 14-18 August, 2007,	Selected presentation	6th International ISAAC Congress, 14-18 August, 2007,
	AIMS International Conference on Dynamical Systems and Differential equations, Poitiers , France, Jun 25-28, 2006,	Selected presentation	AIMS International Conference on Dynamical Systems and Differential equations, Poitiers , France, Jun 25-28, 2006,
	International Conference “Differential and Difference Equations and Applications”, Melbourne, Florida, USA, August 1-5,2005,	Selected presentation	International Conference “Differential and Difference Equations and Applications”, Melbourne, Florida, USA, August 1-5,2005,
	International Conference “Navier – Stokes Equations and Applications”, St.Petersburg, Russia, September 11-18,2002,	Selected presentation	International Conference “Navier – Stokes Equations and Applications”, St.Petersburg, Russia, September 11-18,2002,
	International Conference “Nonlinear Partial Differential Equations”, Alushta, the Crimea, Ukraine, September 15-21,2003,	Selected presentation	International Conference “Nonlinear Partial Differential Equations”, Alushta, the Crimea, Ukraine, September 15-21,2003,

Year	Event, place	Type	Title of talk
	Ninth International Symposium on continuum models and discrete systems, Istanbul, 29 June - 3 July 1998,	Selected presentation	Ninth International Symposium on continuum models and discrete systems, Istanbul, 29 June - 3 July 1998,
	International workshop on turbulence modelling and vortex dynamics, Istanbul, 2-6 September 1996,	Selected presentation	International workshop on turbulence modelling and vortex dynamics, Istanbul, 2-6 September 1996,
	International workshop on non-linear problems in mathematical physics II, Ankara, September 14-18, 1995,	Selected presentation	International workshop on non-linear problems in mathematical physics II, Ankara, September 14-18, 1995,
	International workshop on dynamical systems, Barcelona, Spain, September, 1994,	Selected presentation	International workshop on dynamical systems, Barcelona, Spain, September, 1994,
	International conference on partial differential equations "Equidiff 91" Barcelona, Spain, August 1991, 3 ===== PAGE 4/8 =====	Selected presentation	International conference on partial differential equations "Equidiff 91" Barcelona, Spain, August 1991, 3 ===== PAGE 4/8 =====
	International conference on continuum mechanics and related problems of analysis dedicated to 100th anniversary of academician N.I.Muskhelishvili, Tbilisi, Georgia, June 1991,	Selected presentation	International conference on continuum mechanics and related problems of analysis dedicated to 100th anniversary of academician N.I.Muskhelishvili, Tbilisi, Georgia, June 1991,
	Annual Iranian mathematical conference, Mashhad, Iran, 1991,	Selected presentation	Annual Iranian mathematical conference, Mashhad, Iran, 1991,
	All-union conference on differential equations and mathematical physics dedicated to 80-th anniversary of academician N.N.Bogolyubov, Ternopol, Ukraine, 1990,	Selected presentation	All-union conference on differential equations and mathematical physics dedicated to 80-th anniversary of academician N.N.Bogolyubov, Ternopol, Ukraine, 1990,
	International conference on partial differential equations, Holzhau, Germany, 1988,	Selected presentation	International conference on partial differential equations, Holzhau, Germany, 1988,
	International topological conference, Baku, 1987,	Selected presentation	International topological conference, Baku, 1987,
	All-union conference on the theory of functions and differential equations dedicated to 100th anniversary of academician V.I. Smirnov, Leningrad, 1987,	Selected presentation	All-union conference on the theory of functions and differential equations dedicated to 100th anniversary of academician V.I. Smirnov, Leningrad, 1987,
	All-union seminar on nonlinear problems of mathematical physics, Leningrad, 1984,	Selected presentation	All-union seminar on nonlinear problems of mathematical physics, Leningrad, 1984,
	All-union conference on mathematical problems of physics, Moscow, 1984,	Selected presentation	All-union conference on mathematical problems of physics, Moscow, 1984,
	All-union conference on nonlinear problems of mathematical physics, Leningrad, 1982,	Selected presentation	All-union conference on nonlinear problems of mathematical physics, Leningrad, 1982,
	All-union conference on mathematical problems of physics, Moscow, 1978,	Selected presentation	All-union conference on mathematical problems of physics, Moscow, 1978,

Year	Event, place	Type	Title of talk
	International conference on partial differential equations dedicated to 75th anniversary of academician I.G. Petrovski, Moscow, 1975.	Selected presentation	International conference on partial differential equations dedicated to 75th anniversary of academician I.G. Petrovski, Moscow, 1975.

Languages

Language	Level
Azerbaijani	Native
English	Fluent
Russian	Fluent
Turkish	Fluent
German	Advanced